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Comparison of measurement techniques and sorption of radium-226 in low and high salinity aqueous samples

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ABSTRACT

Human activities have the potential to redistribute radium (Ra) in the marine environment in a manner that may necessitate monitoring or management of subsequent human or environmental exposures. There is therefore a need to identify accurate and accessible techniques for Ra measurement in high salinity samples and to describe the distribution of Ra in estuarine and marine environments, but most efforts in these areas have focused on low salinity matrices. In addition, rapid and reliable measurements are crucial for time-sensitive samples such as short-lived isotopes or emergency situations. The objective of this study is to describe the limits of detection, cost, and relative ease for measurement of Ra in both low and high salinity aqueous samples via three analytical methods: liquid scintillation counting (LSC), high purity germanium (HPGe) gamma spectrometry, and inductively coupled plasma mass spectrometry (ICP-MS). To contextualize these measurements for real-world scenarios, the partitioning of ^{226}Ra to substrates relevant to the marine environment was also characterized. Although HPGe detection with solid phase extraction had the lowest limit of detection for low salinity samples (0.27 Bq L^{-1}), poor ^{226}Ra recovery for high salinity samples and high materials costs make this method prohibitive for many users. Limits of detection for high salinity samples were lower for LSC (1.28 Bq L^{-1}) than for ICP-MS without dilution (11.4 Bq L^{-1}), but significant and unexpected degradation of the high salinity LSC standards was observed after six months. Therefore, our preferred measurement method for high salinity Ra samples is ICP-MS with sample dilution as necessary to reduce matrix effects.

1. Introduction

Human activities have the potential to relocate and concentrate naturally occurring radioactive material (NORM) such that environmental monitoring or remediation is needed to meet regulatory limits or ensure protection of human health. NORM that has been disturbed or otherwise altered by human activities is often referred to as NORM residues (IAEA, 2013a, 2013b) or technologically enhanced NORM (TENORM) (US EPA, 2008) to distinguish from unimpacted NORM. For example, radium (Ra) is a naturally occurring radionuclide found in elevated activity concentrations (worldwide average of ~1 Bq/g) in phosphogypsum, a waste byproduct of fertilizer production (IAEA, 2013b). Concerns surrounding phosphogypsum stacks, or waste ponds, include the potential for increased radon (Rn) concentrations in the local area as Rn is a gaseous decay product of Ra (IAEA, 2013b). In the United States, Rn emissions from phosphogypsum stacks are regulated under the Clean Air Act. NORM residues are also of concern in the oil and gas industry, including Ra in high salinity wastewater (~500–700 Bq L^{-1}) associated with unconventional drilling known as "fracking" (Thakur et al., 2021) and buildup of Ra (~40–1000 Bq/g) in scale of offshore petroleum pipelines (e.g., MacIntosh et al., 2022). The concern with the former is the potential for elevated Rn concentrations and potential for Ra to migrate into the human food chain; the concern with the latter is the potential for direct exposure of marine organisms.

Exposures to workers, members of the public, and the natural environment from industrial activities involving NORM or NORM residues is not necessarily the dominant risk associated with the activity, and a graded approach to the corresponding management should be integrated and consistent with the management of other hazards (ICRP, 2019). There are circumstances, such as those mentioned above, where

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control of longer-term exposure may be necessary. In these cases, robust, sensitive, and practical measurement techniques for environmental samples are needed to ensure that protection standards are being appropriately met, both for human health and environmental protection. Similarly, high quality measurements can inform protection standards by ideally reducing uncertainties.

Although several recent reviews have described best practices for Ra sample preparation, separation, and measurement in environmental samples (Abbasi, 2018; Boudias et al., 2022; Diego-Feliu et al., 2020; Jia and Jia, 2012; Thakur et al., 2021), the analysis and discussion of these techniques for high salinity samples are less common (but see Henderson et al., 2013; Song et al., 2017; van Beek et al., 2010). Jia and Jia (2012) describe various separation and measurement techniques of Ra in environmental samples and found that alpha (α) spectrometry had a significantly lower minimum detectable activity (MDA) compared to gamma (γ) spectrometry or liquid scintillation counting (LSC), and was therefore a better choice for low-level counting of ^{226}Ra. Thakur et al. (2021) also reviewed methods of sample preparation for measuring Ra isotopes in environmental samples and note that recent advances in mass spectrometry now allow for measurement of ultra-trace levels of Ra, making it ideal for environmental samples. Although neither of these reviews directly compared low vs high saline water samples, Thakur et al. (2021) note that evaporation (a common pre-concentration technique for measuring Ra in liquid samples) is typically not feasible for seawater samples.

Typical pre-concentration methods for Ra in both high and low salinity solutions include co-precipitation and adsorption (Thakur et al., 2021). Co-precipitation is also a convenient separations technique and often is done with barium sulfate ($BaSO_4$) as it has a high selectivity for Ra (Jia and Jia, 2012). However, reagent-grade $BaSO_4$ contains trace but variable amounts of ^{226}Ra, therefore reducing the sensitivity and accuracy of the measurements (Thakur et al., 2021). Adsorption of Ra onto manganese dioxide (MnO_2) beads, coated fibers, or cartridges is also a common preconcentration technique. However, this method can be time-consuming, inaccurate, or inefficient (Jia and Jia, 2012). As high salinity matrices can require more intensive pre-concentration techniques (e.g., Song et al., 2017), it is useful to identify measurement methodologies with the lowest limits of detection that also limit the need for high volume samples (e.g., 100–1000s of L) (Henderson et al., 2013) or excessively cumbersome or lengthy sample pre-processing. The high salt content of marine waters typically contributes to a complex sample matrix that is challenging for many analytical approaches.

This study directly compares rapid turnaround measurements of Ra in low vs high saline water samples via three detection methods characterized by comparatively simple sample preparation and high measurement accuracy: LSC, γ-spectrometry via HPGe, and single quadrupole inductively coupled plasma mass spectrometry (ICP-MS). We describe the detection limits, ease, and cost of each method to help researchers determine which method best suits their samples to minimize sample preparation in situations when rapid turnaround is required. Additionally, because Ra behavior differs between freshwater and marine systems, we conducted batch sorption experiments to calculate the K_d values of ^{226}Ra to various solids relevant to the marine environment in both high and low salinity solutions. We hypothesized that higher salinity would enhance Ra desorption from sediments and solids in water and result in a lower K_d value, while the lower ionic strength of freshwater solutions would contribute to a greater fraction of bound Ra and a higher K_d value. The results herein are intended to facilitate accurate, rapid, and reliable methods for Ra analysis in high salinity systems which support appropriate assessment and management of environmental exposure to Ra.

2. Materials and methods

2.1. Radium-226 standards

Freshwater (0 PSU) and saltwater (35 PSU) ^{226}Ra standards for the LSC and ICP-MS instruments were created with nominal activity concentrations ranging from 37 to 185,000 Bq kg^{-1} (0.001–5 μg kg^{-1}; assuming 1 μg ^{226}Ra = 1 μCi ^{226}Ra) in 2 % HNO_3. A 3970.1 kBq kg^{-1} (107.3 μg kg^{-1}) ^{226}Ra working solution in 2 % HNO_3 was first prepared from a 185 kBq mL^{-1} (5 μg mL^{-1}) ^{226}Ra stock solution in 1 M HNO_3 (Eckert & Ziegler Isotope Products). The exact concentration of ^{226}Ra in the working solution was confirmed via a HPGe detector (Ortec) with an efficiency calibration based on a NIST traceable ^{152}Eu source. Freshwater standards were diluted from the stock solution in distilled deionized (DDI) water to a measured mass/activity concentration range of 0.0001–5.4 μg L^{-1} (3.7–199,800 Bq L^{-1}). Saltwater standards were diluted from the stock solution in 35 PSU synthetic seawater to a measured mass/activity concentration range of 0.0001–5.3 μg L^{-1} (3.7–196,100 Bq L^{-1}). Synthetic seawater at 35 PSU (equivalent to 35 ppt) was created using Instant Ocean® Sea Salt according to the manufacturer's instructions by adding 42 g of sea salt to 1 L of DDI water. Salinity was checked with a salinity refractometer with automatic temperature compensation with a resolution of 1 ppt (VEE GEE Scientific, LLC). All standards were pH adjusted to ~1 to limit analyte sorption onto container walls using 2 % HNO_3.

2.2. Liquid scintillation counting (LSC)

To prepare the standards for liquid scintillation counting (LSC), 10 mL of each standard and 10 mL of ultra-low level liquid scintillation mineral oil (PerkinElmer Ultima Gold™ LLT) was transferred to a glass LSC vial. Using this method, radon-222 (^{222}Rn) is extracted from the sample into the mineral oil and the ^{226}Ra activity is measured indirectly through the activity of the equilibrium progeny, specifically ^{222}Rn (IAEA, 2010; Jia and Jia, 2012). The primary drawback of this method for environmental samples is the non-specificity for the target analyte as the count will include other naturally occurring radionuclides such as those in the U and Th series (Jia and Jia, 2012). The samples must be sealed and allowed to sit for 30 days for the ingrowth of ^{222}Rn to reach secular equilibrium with its parent ^{226}Ra. After 30 days, our sealed standards were vigorously mixed via inversion for 60 s, and left undisturbed for at least 4 h until counting. The standards were counted on two instruments of varying sensitivity, a PerkinElmer Tri-carb 4910 TR and a Quantulus 1220 Ultra. Additional instrument and counting details are available in the Supporting Information. After counting, standards were allowed to sit on a lab bench for six months and then were recounted on the PerkinElmer Tri-carb 4910 TR to detect any difference in quench between the two sets of standards over time.

The background count rate within the ^{226}Ra region of interest (ROI) was used to determine the limit of detection employing the Currie equation for equal sample and background count times (Currie, 1968):

$$\mathrm{LLD} = 2.71 + 4.65\sqrt{C_{T,\mathrm{ROI}}} \quad \text{[Eqn 1]}$$

where LLD is the limit of detection and $C_{T,ROI}$ is the total number of counts within the region of interest (ROI) from the background measurement. From the LLD, the detection limit can be determined as:

$$\mathrm{MDC} = \frac{\mathrm{LLD}}{V \times t \times \varepsilon(E) \times f_e \times Y} \quad \text{[Eqn 2]}$$

where MDC is the minimum detectable activity concentration (Bq/L), V is the sample volume (L), t is the measurement time (s), $\varepsilon(E)$ is the energy (E) dependent detection efficiency (i.e., for ^{226}Ra 186.7 keV gamma ray) determined from the ratio of the measured disintegrations per minute (DPM) to the actual DPM for LSC and from the detection efficiency curve for HPGe (see Section 2.3), f_e is the ^{226}Ra gamma emission fraction, and

Y is the chemical yield (recovery) for any separations process. Uncertainty calculations are described in more detail in the Supporting Information.

2.3. High purity germanium (HPGe)

Two approaches were considered for determination of detection limits for ^{226}Ra in liquid samples using high resolution, high purity germanium (HPGe) detectors: (1) direct measurement of 250 mL of solution and (2) solid phase extraction of 1000 mL of solution onto specialized Solid Phase Extraction (SPE) filter disks (CDS Analytical 3291 Empore™ Radium SPE RAD Disk). In both cases a NIST traceable ^{152}Eu source in 0.502 mL of 2 % nitric acid (Eckert and Zeigler Isotope Products, Atlanta, GA) was used to determine HPGe efficiency curves. A ^{152}Eu standard was used due to the lack of availability of a similarly traceable ^{226}Ra standard. A Canberra GC 4018 HPGe cooled using a Cryo-cycle interfacing with a Lynx multichannel analyzer and Prospect software was used for all measurements.

In the first direct measurement scenario, a 250 mL DDI water sample was spiked with 252 Bq (6811 pCi) of the ^{152}Eu source and then measured directly for 24 h. The background count rate was determined by taking a 24-h measurement with no sample within the detector. Measurement data was used for energy calibration and determination of detector efficiency (Figure S1A). The ^{226}Ra peak ROI of 184.7–187.8 keV was used to define the background region of the ^{152}Eu spectra and the LLD and MDC for ^{226}Ra were calculated as described for the LSC method. Defining the background region as that of the ^{226}Ra gamma of interest enables determination of LLD and MDC.

Note that this measurement was only completed for freshwater but would be consistent with that in saltwater. Although photointeractions are influenced by material density and composition, 186 keV photons are of high enough energy to be unimpacted by the subtle differences associated with level of salinity, particularly at the spatial scale of interest. For example, total attenuation for water with no salt compared to water with 35 ‰ salt (i.e., NaCl) for 186 keV photons is 0.1404 cm^2/g compared to 0.1400 cm^2/g, respectively (NIST XCOM).

To determine detector efficiency for the SPE filter disk scenario, a 12.64 Bq (341.6 pCi) aliquot of ^{152}Eu solution was added to a glass fiber filter adhered with tape to a piece of clear packaging tape. The packaging tape was included to prevent any of the solution from going onto the bench or absorbent paper while the sample was drying. After drying overnight, a second filter paper was placed on top of the ^{152}Eu amended filter and both were sealed with an additional layer of packing tape. The assembled filters were then placed in a 2" diameter Petri dish which was then counted for 24 h on the HPGe.

The ^{226}Ra recovery for low ionic strength solutions was assumed to be quantitative (≥95 %) during SPE filtration according to manufacturer's instructions, but typical saltwater ion concentrations exceeded the recommended levels (Table S1; 3 M, 2000). Therefore, to determine ^{226}Ra recovery for typical seawater applications, two filtration replicates were extracted according to the manufacturer's instructions (3 M, 2000). Briefly, 225 Bq (6.08 nCi) of ^{226}Ra was added from a 3970.1 kBq kg^{-1} (107.3 μg kg^{-1}) ^{226}Ra working solution in 2 % HNO_3 to 1 L of 35 PSU synthetic seawater and the sample was acidified to 1 M with concentrated HNO_3. An Empore™ SPE disk was carefully placed on a vacuum filtration system which was maintained at an approximate flow rate of 50 mL/min throughout the rest of the process. The SPE disk was first washed with approximately 20 mL of 2 M HNO_3 and then the sample was immediately added. After the entire sample was filtered, the SPE disk was once again rinsed with approximately 20 mL of 2 M HNO_3 followed by an additional drying rinse of 20 mL of acetone. The disk was then dried at 60 °C on a metal planchet for 30 min before being sealed in a 55 mm × 60 mm mylar bag. The bag was directly placed onto the shielded HPGe detector and counted for 24 h. The activity on the filter was calculated by dividing the measured activity by the detection efficiency previously determined from the simulated freshwater filter. The ^{226}Ra recovery was calculated by dividing the filter activity by the total activity added to each sample. Finally, the MDC was calculated as previously described in Equation (2), with the ^{226}Ra recovery as the yield in the denominator of the equation.

2.4. Inductively coupled plasma mass spectrometry (ICP-MS)

No additional sample preparation was done prior to measurement of the previously described freshwater and saltwater standards via ICP-MS (Section 2.1). Both sets of standards were analyzed on a Thermo-Scientific iCAP RQ ICP-MS during the same run. Samples and standards were run on KED (kinetic energy discrimination) mode to suppress interfering polynuclear complexes with ^{209}Bi as an internal standard. The calibration curve was reported automatically by the Qtegra software, and we fit separate blank-forced, non-weighted linear curves to the freshwater and saltwater standards. The limit of detection (LOD) for ICP-MS is referred to as the instrument detection limit (IDL), (μg L^{-1}) and was calculated as:

$$\mathrm{IDL} = C_{\mathrm{STD}}\left(\frac{3\ \mathrm{SD}_{\mathrm{blank}}}{\mathrm{intensity}_{\mathrm{STD}} - \mathrm{intensity}_{\mathrm{blank}}}\right) \quad \text{[Eqn 3]}$$

where $\mathrm{SD}_{\mathrm{blank}}$ is the standard deviation the blank (cps), C_{STD} is the concentration of the standard (μg L^{-1}), $\mathrm{intensity}_{\mathrm{STD}}$ and $\mathrm{intensity}_{\mathrm{blank}}$ are the average intensities (cps) of the standard and blank respectively (Agilent Technologies, 2005). The background equivalent concentration (BEC), which describes the magnitude of the blank signal, is calculated within the Qtegra software as Equation (4):

$$\mathrm{BEC} = C_{\mathrm{STD}}\left(\frac{\mathrm{intensity}_{\mathrm{blank}}}{\mathrm{intensity}_{\mathrm{STD}} - \mathrm{intensity}_{\mathrm{blank}}}\right) \quad \text{[Eqn 4]}$$

2.5. Batch sorption experiment

Batch sorption experiments were conducted to calculate the K_d values of ^{226}Ra to various solids in both artificial (Instant Ocean® Sea Salt) 600 mM saltwater and 10 mM NaCl solutions. Four solids were chosen as adsorbents: quartz, crushed *Geukensia demissa* (Atlantic ribbed mussel) shell primarily consisting of calcite, a montmorillonite clay, and iron oxide (goethite). Either 0.25 g (quartz, crushed shell, and montmorillonite) or 0.05 g (goethite) of the adsorbent was added to a preweighed 15 mL centrifuge tube and then reweighed. A lower goethite mass was used compared to other solids due to the relatively large surface area of goethite. A specified volume of either artificial seawater or 10 mM NaCl solution was then pipetted into the tube followed by addition of ^{226}Ra at nominal concentrations of 185, 925, and 1850 Bq L^{-1} (5, 25, and 50 ng L^{-1}; Table 1). The total liquid volume of the solution was fixed to 10 mL and all samples were prepared in duplicate.

Samples were placed on an end-over-end shaker to mix for 24 h to allow sufficient time for sorption equilibrium. After, samples were removed from the shaker and centrifuged for 20 min at 4500 rpm to remove particles with 100 nm or greater diameter from the solution. One mL of supernatant was pipetted into a new 20 mL glass vial and 10 mL of mineral oil were added to each sample, with exact volumes determined gravimetrically. The remaining suspensions were returned to the shaker for another 7 days for a future equilibrium check to ensure the supernatant samples were at steady state. Samples were then

Table 1
Composition and concentrations of the solutions used for the batch sorption experiments.

Target ^{226}Ra Concentration (Bq L^{-1})	Volume of 37 kBq L^{-1} ^{226}Ra Working Solution (mL)	Volume of Seawater or NaCl Solution (mL)
185	0.05	9.95
925	0.25	9.75
1850	0.5	9.5

measured via LSC as described before. A mass balance of the measured aqueous ^{226}Ra concentrations and known total ^{226}Ra concentrations was applied to calculate the solid phase ^{226}Ra concentrations. The solid/liquid equilibrium partitioning coefficient (K_d) for each material was calculated as:

$$K_d = \frac{C_s}{C_w} \quad \text{[Eqn 5]}$$

where C_s is the solid phase ^{226}Ra concentration and C_w is the ^{226}Ra measured aqueous concentration. A one-way analysis of variance (ANOVA) test with solution type as an explanatory factor were run for each material at the 0.05 significance level in JMP.

3. Results and discussion

3.1. Liquid scintillation counting

The Tri-carb and Quantulus LSC instruments both recorded higher counts for the saltwater standards than freshwater standards of the same ^{226}Ra concentration (Fig. 1A). The detection limit for LSC is a function of counting time, sample volume, and instrument sensitivity, with previous studies finding detection limits around 0.05–20 Bq L^{-1} (1.35–540.4 pCi L^{-1}) for Ra (Jia and Jia, 2012). Our results are in good agreement, with detection limits ranging from 0.07 to 1.28 Bq L^{-1} (1.89–34.6 pCi L^{-1}) for one-month old samples (Tables 2 and 3). The MDC for the saltwater standards was marginally higher than the freshwater standards for both instruments, although at higher concentrations a greater difference between counts was observed compared to lower concentrations (Table 2; Fig. 1A). The Tri-carb instrument MDC was ~13× greater than the Quantulus instrument for both fresh- and saltwater standards, indicating lower sensitivity by approximately an order of magnitude (Table 2). Ultimately, the salinity did not greatly impact the detection of ^{226}Ra in freshly made samples via LSC. However, for the six-month old standards, we observed a minor increase (28 %) in the MDC for the freshwater standards (i.e., only a small increase in quench) but the MDC for the saltwater standards more than doubled, from 1.28 Bq L^{-1} to 2.71 Bq L^{-1} (1.89–73.2 pCi L^{-1}) (Table 4; Fig. 1B). This was attributed to significant change in the coloration of the saltwater standards over six months (Fig. 2).

In LSC, the sample is combined with solvent and a scintillator ("fluor") to stimulate production of photons which are quantified to determine the sample activity. This method is ideal for low energy beta- and alpha-emitters. The emitted energy excites solvent molecules and is then transferred to the scintillator which produces photons detected by a photoemitter tube (PMT). The primary advantages of this technique are the relative ease of sample preparation, ability to measure high sample volume, and high counting efficiency up to 100 % (Knoll, 2010). The drawbacks include large instrument sizes, poor resolution contributing to difficulties in the identification of radionuclides, reductions in efficiency due to chemical or color quenching, and the hazardous nature of some of the chemicals used for the fluor component (Knoll, 2010). Commonly, LSC measurement of ^{226}Ra activity is quantified from the decay products, so the long wait time to reach secular equilibrium (~30 days) in the sealed sample is an additional drawback specific to this radionuclide. In this study, the interlinked consequences associated with this decay time and the associated color quenching were significantly impactful.

To the best of our knowledge, the significant sample discoloration observed in this study has not been previously reported but represent a significant limitation for this method with saltwater samples. Although the mechanisms for this color change are currently unknown, the severity of the reaction over time is variable among samples but appears to progress over the course of multiple years in samples with salinity ≥20 PSU (Figure S2). Defining the underlying mechanisms behind this degradation and identifying how to either stop or slow the associated color quenching represents an interesting area for future study. Prior studies have noted variations in quench resistance among LSC cocktails in samples stored on timescales of months to years (Feng et al., 2015), indicating that thoughtful selection of cocktail is critical for samples that either require an ingrowth period or long storage time. The LSC cocktail used in this study, Ultima Gold™ LLT, has been shown to be an average to above average performer when compared among nine commercially-available LSC cocktails in terms of ^{3}H/^{14}C counting efficiency, quench resistance, and sample stability, although some evidence suggested that salinity decreased this performance (Verrezen et al., 2008). However, there are generally few direct comparisons available between low and high salinity samples measured via LSC, although some studies have reported increased quenching in high salinity samples (Palomo et al., 2011).

A Ra-specific LSC cocktail RADAEX® has previously been developed which contains 2-methyl-2-2-heptylnonanoic acid (HMHN) and dicyclohaxano-21-crown-7 ($Cy_2$21Cy) as the solvent extractants (McDowell et al., 1989). However, this extraction necessitates an alkaline pH between 10 and 12, which results in significant salt

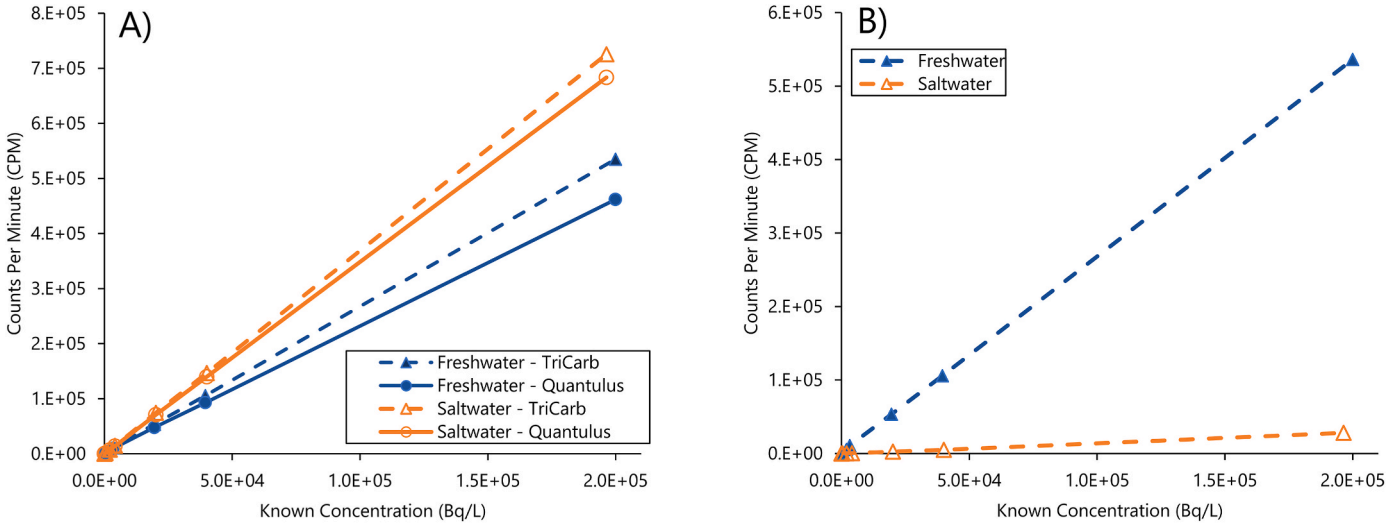


Fig. 1. Counts per minute for freshwater (blue, closed markers) and saltwater (orange, open markers) standards after (A) one month on the Tri-carb (dashed lines) and Quantulus (solid lines) instruments and (B) 6 months on the Tri-carb instrument. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

Table 2
LSC counting statistics for one-month-old ^{226}Ra mineral oil standards.

Instrument	Standard Type	LLD (Counts)	Efficiency (%)	MDC (pCi L^{-1})	MDC (Bq L^{-1})
Tri-carb	Freshwater	236 ± 15.3	587 ± 18.4	30.2 ± 2.0	1.12 ± 0.07
	Saltwater	270 ± 16.3	585 ± 24.5	34.6 ± 2.1	1.28 ± 0.08
Quantulus	Freshwater	171 ± 13.0	423 ± 11.1	1.90 ± 0.14	0.070 ± 0.005
	Saltwater	286 ± 60.9	463 ± 13.1	2.90 ± 0.17	0.107 ± 0.006

Table 3
Summary of the limits of fresh- and salt-water limits of detection and ease/cost for the three methods evaluated for ^{226}Ra detection.

Detection method	Freshwater detection limit (Bq L^{-1})	Saltwater detection limit (Bq L^{-1})	Ease of method	Cost of method
LSC[a] (10 mL)	1.12	1.28[c]	simple	low
HPGe (250 mL direct counting)	6.29	6.29[b]	simple	low
HPGe (1 L Empore™ SPE disk filtration)	0.27	5.86	moderate	high
ICP-MS (no dilution)	0.41	11.4	moderate	moderate[d]

[a] Tri-carb 4910-TR instrument; detection limits can be lower for more sensitive LSC.
[b] Based on freshwater detection limit.
[c] Based on a 30-day old sample, longer time frames may increase detection limits due to sample degradation.
[d] Cost of purchasing NIST standard for calibration is moderately high ($1800), consumables are reasonable.

precipitation, making this method unsuitable for high salinity water samples (Hou, 2018). High salinity environmental samples are vulnerable to a variety of mechanisms for quenching beyond the discoloration observed in this study, such as the aforementioned precipitation of salt or presence of other suspended solids which can cause physical quenching, or chemical quenching caused by certain molecules present in seawater such as dissolved oxygen or organic acids (Schubert and Kallmeyer, 2023). For the discolored high salinity samples in this study, potential but unverified mitigation strategies include decoloration via ultraviolet radiation (UV) exposure or addition of a small volume of 30 % hydrogen peroxide (H_2O_2) to bleach the sample (US NRC, 2011). Note that opening the samples to add H_2O_2 would require waiting for the 30-day ingrowth period again prior to measuring. Although the aim of this study was to compare detection methods characterized by comparatively simple sample preparation, future studies should explore if additional sample pre-treatment such as filtration or addition of a pH buffer reduces the observed discoloration in the samples. Therefore, LSC may be a convenient method for Ra detection in environmental saltwater samples, but the user must remain cognizant of the impacts to measurement from possible discoloration or degradation of the solvent or fluor components over time.

3.2. High purity germanium (HPGe)

The detection efficiency curves for the 250 mL water sample and SPE

Table 4
LSC counting statistics for six-month-old ^{226}Ra mineral oil standards.

Instrument	Standard Type	LLD (Counts)	Efficiency (%)	MDC (pCi L^{-1})	MDC (Bq L^{-1})
Tri-carb	Freshwater	236 ± 15.3	456 ± 9.6	38.9 ± 2.5	1.44 ± 0.09
	Saltwater	21.8 ± 4.4	22.4 ± 3.5	73.4 ± 18.6	2.71 ± 0.70

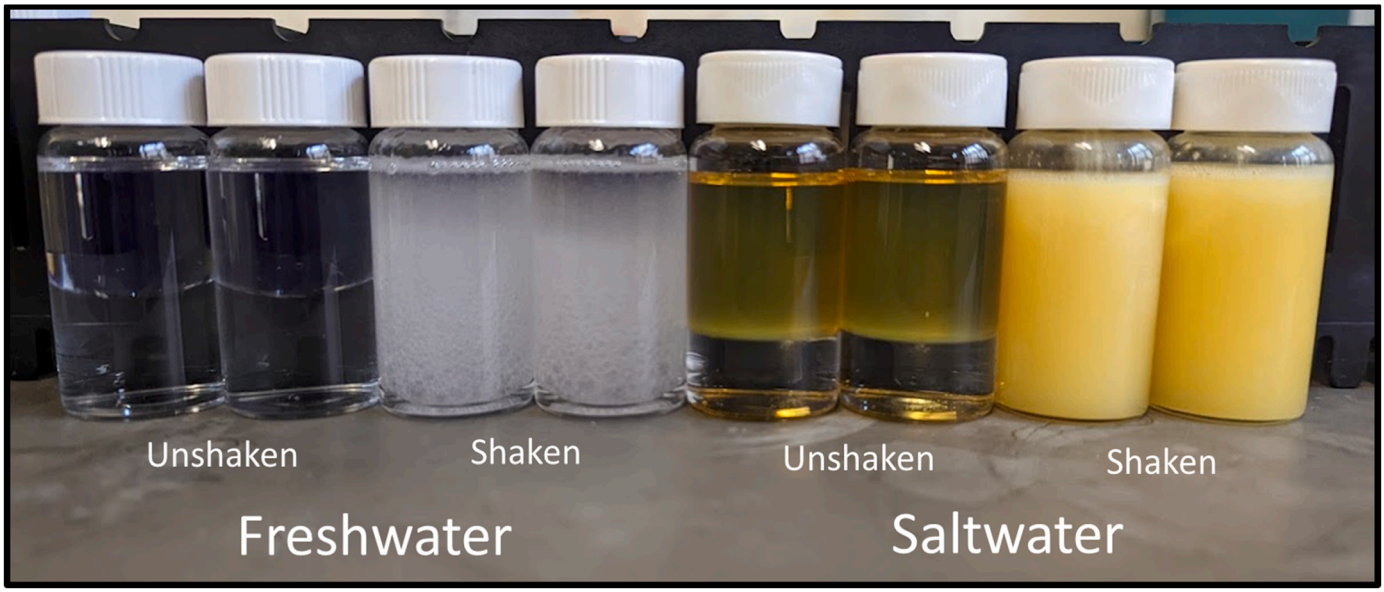


Fig. 2. Image of the six-month-old LSC mineral oil ^{226}Ra standards showing the yellow discoloration of the saltwater standards. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

disk are provided in Figure S1 and indicate detection efficiencies of 0.023 and 0.011, respectively. The freshwater MDCs for the two approaches were 6.29 Bq L^{-1} (170 pCi L^{-1}) for direct counting of a 250 mL water sample and 0.27 Bq L^{-1} (7.4 pCi L^{-1}) for extraction of 1 L of water through the SPE disk (Table 3). For saltwater samples, the poor ^{226}Ra recovery yield for the SPE disks was 4.7 ± 0.4 % resulted in a significantly higher MDC of 5.87 Bq L^{-1}.

HPGe detectors, a type of semiconductor detector, are used for a variety of applications including environmental monitoring, nuclear forensics, and threat reduction (Casagrande and Bonotto, 2018; US EPA, 2019; Ramebäck et al., 2023). As with LSC, we were unable to identify any prior studies reporting direct comparisons between low and high salinity samples, but HPGe has effectively been used to determine activity in high salinity samples (Grabowski and Bem, 2010) and even for *in situ* underwater measurements in seawater (Povinec et al., 1996). In this study, the direct counting of a water sample had a notably higher (23.3 ×) detection limit for the freshwater samples and marginally higher (1.07 ×) detection limit for the saltwater samples compared to extraction on SPE disks, but is a very simple and inexpensive approach compared to the expensive SPE disks. However, both methods require a large sample volume and relatively long counting time, which may make HPGe detection infeasible in cases of low sample amounts and/or high numbers of independent samples.

Note that another concern when measuring ^{226}Ra in seawater with HPGe is the potential interference of natural uranium, which has a concentration in 35 ‰ seawater of about 3.3 μg/L (Ku et al., 1977). Although our synthetic seawater standards did not contain uranium, uranium-235 (0.72 % of naturally occurring uranium) emits a gamma of 185 keV in its decay (ICRP, 2008) and therefore interferes with the ability to clearly distinguish the ^{226}Ra peak in the gamma spectrum. ^{226}Ra and its progeny have additional gammas that can be measured to disambiguate from ^{235}U (e.g., the gamma emission of the progeny isotope ^{214}Bi at 1764 keV), but samples would need to meet the conditions for being in secular equilibrium to do so.

We also report here an unsatisfactory ^{226}Ra recovery yield of <5 % for the SPE disks, likely due to the large seawater sample volume measured (1 L). Lagacé et al. (2017) also noted a relationship between sample volume and Ra recovery from high salinity samples onto a strong cation exchange resin (AG50W-X8), with recoveries decreasing from 99 ± 4 % for 10 mL samples of natural seawater to 57 ± 8 % for 90 mL samples. This was attributed to increased mass of salts loaded onto the resin. The recovery of ^{226}Ra in high salinity fracking wastewater for the Empore™ SPE disks has also previously been reported at 13 ± 1 % (Nelson et al., 2014). Taken together with our results, the previously reported low recoveries for high salinity environmental samples suggest that the use of SPE disks is extremely limited for high ionic strength solutions.

3.3. Inductively coupled plasma mass spectrometry (ICP-MS)

Reported ICP-MS intensities were 25 ± 8.4 × lower for saltwater standards than for freshwater standards (Fig. 3). The LOD for the freshwater and seawater standards were 0.41 Bq L^{-1} (11.14 pCi L^{-1}) and 11.4 Bq L^{-1} (308.5 pCi L^{-1}), respectively (Table 5). Significant scatter due to signal noise for the low concentration standards can contribute to an excessively high reported LOD, particularly for the high salinity standards (Fig. 3). This can be avoided by either weighting the regression by the relative standard deviation or, as was done in this study, forcing the curve through the blank. As these approaches resulted in essentially identical LODs for the seawater standards in this study, either would be acceptable to avoid over-emphasis of the high concentration standards in the calibration curve.

In contrast to LSC and HPGe, ICP-MS is not specifically a radiation detection technique, but rather an analytical technique capable of measuring a wide variety of elements at trace levels in a liquid sample (Wilschefski and Baxter, 2019). The primary benefits of ICP-MS are the ability to measure multiple elements simultaneously, low detection limits for most elements, low required sample volume, and ease of sample preparation. The primary drawbacks are the large initial instrument costs, challenge of resolving isotopic interferences, and the laboratory expertise needed to maintain and interpret the instrument (Wilschefski and Baxter, 2019). The challenges involved with measuring high salinity samples via ICP-MS are primarily associated with reduced instrument precision and sensitivity due to the presence of easily ionized elements like sodium, salt deposition on sensitive internal components, and formation of polyatomic ions within the plasma (Sun et al., 2018). As the recommended total dissolved solids (TDS) concentration for ICP-MS is very low (<0.2 %) (Thomas, 2004), dilution will likely be necessary for measuring most elements in high salinity samples.

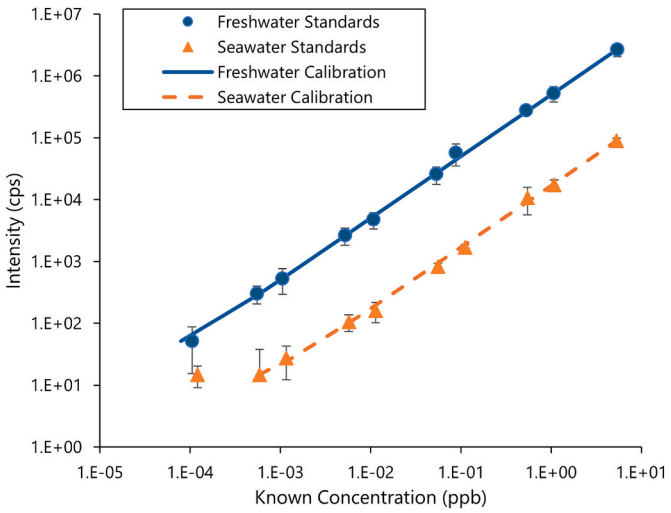


Fig. 3. Intensity (counts per second) values and calibration curves for the freshwater (blue) and saltwater (orange) standards on the ThermoScientific iCAP RQ ICP-MS. Error bars represent the standard deviation of five consecutive measurements. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

For example, our results demonstrate a potential need for sample dilution prior to analysis, which reduces the concentration of ^{226}Ra within the sample but simultaneously improves the overall sensitivity of the instrument by limiting the number of competing ions in solution and lowering the detection limit. ICP-MS is also advantageous over both LSC and HPGe when rapid turnaround is desired or a high number of total samples are involved, as there is no associated waiting time for progeny ingrowth as with LSC and counting times for individual samples are on the order of minutes, rather than hours as with HPGe.

3.4. Batch sorption

To better contextualize the measurement of ^{226}Ra in high salinity aqueous environmental samples, it is necessary to develop an improved understanding of the partitioning of ^{226}Ra between the solid and aqueous phase in estuarine and marine environments. We conducted batch sorption experiments on a wide range of pure mineral phase which represent minerals commonly found in estuarine and marine environments. These solids included the most common crystalline mineral found in sand to represent silicate (quartz), a biogenic material commonly found in estuaries (calcium carbonate shell), a fine-grained sediment which is commonly found in shallow marine sediments (montmorillonite clay), and an iron oxide found in marine sediments (goethite). The impacts of seawater on ^{226}Ra partitioning to these pure mineral phases indicates the importance of performing measurements under saline conditions which necessitates the more detailed understanding of the analytical capabilities and limitations presented in the current work.

Table 5
Results from the linear regression for the ICP-MS limit of detection.

Standard Type	Regression	Coefficients	Standard Error	BEC (pCi L^{-1})	BEC (Bq L^{-1})	LOD (pCi L^{-1})	LOD (Bq L^{-1})
Fresh-water	Intercept	1.33	N/A[a]	2.71	0.10	11.14	0.41
	Slope	491503	1202				
Salt-water	Intercept	1.33	N/A[a]	75.1	2.78	308.5	11.4
	Slope	17755	115.6				

[a] Standard error for the intercept is not available as fit was forced through the blank.

In support of our hypothesis, we observed a significant reduction of Ra sorption to montmorillonite and crushed mussel shells in the high salinity media compared to the low salinity media (Fig. 4). The calculated K_d for crushed shell and montmorillonite clay were 50 × and 9 × higher, respectively, for 10 mM NaCl solution than the 600 mM artificial seawater solution (crushed shell, $p = 0.07$; clay, $p = 0.04$; Fig. 4). Interestingly, the K_d values for iron oxide (goethite) in both solutions were nearly identical ($p = 0.99$). This discrepancy may be a manifestation of the sorption mechanism, with surface complexation by terminal oxygen groups dominating Ra sorption to iron oxide and ion exchange dominating Ra sorption to clay and the crushed shells.

Our hypothesis of reduced sorption in high salinity solution also did not prove accurate for the quartz substrate. Quartz demonstrated a 2 × increase in the K_d for the seawater solution compared to the NaCl solution ($p = 0.004$; Fig. 4). Significant variations among substrates were also noted. The crushed shell solid had a K_d ~3 orders of magnitude higher than quartz for the 10 mM NaCl solution but was approximately equivalent between the two substrates for the 600 mM artificial seawater solution, indicating that competition of the other cations in seawater were similar for these materials.

The influence of temperature on ^{226}Ra sorption to montmorillonite clay have previously been described in a 10 mM NaCl solution (Ames et al., 1983). Specifically, the K_d was found to be 3200 kg L^{-1} at 5 °C, 3450 kg L^{-1} at 25 °C, and 1140 kg L^{-1} at 65 °C (Ames et al., 1983). In contrast, the room temperature (~20 °C) montmorillonite clay K_d in this study was 1691 kg L^{-1}, approximately half of the previously reported value from Ames et al. (1983). The discrepancy between these two measurements is potentially attributed to different sources of montmorillonite used in each study. Prior studies have also examined the sorption of Ra to iron oxides. The K_d of Ra to goethite in a 0.1 M $NaClO_4$ solution increased from 27 L kg^{-1} at a pH of 6–188,000 L kg^{-1} at a pH of 9 (Sajih et al., 2014). At the pH most similar to our experiment (~7), the previously reported K_d was 282 L kg^{-1} (Sajih et al., 2014), comparable to our value of 437 ± 100 L kg^{-1}.

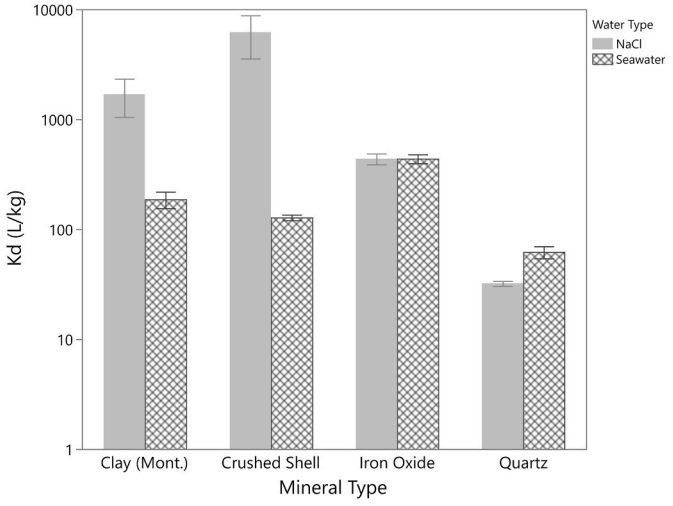


Fig. 4. ^{226}Ra K_d values in a 10 mM NaCl (solid bar) and 600 mM 35 PSU (grid bar) saltwater solution for various solids (n = 6). Brackets with asterisks indicate significant differences between the solution types at the 0.05 significance level. Error bars indicate replicate standard error.

An increase in salinity is typically associated with a decrease in total metal sorption capacity to various solids (Acosta et al., 2011; Esfandiar et al., 2022). This is broadly attributed to higher salinity promoting the desorption due to the presence of competing ions. This competition is expected to be particularly influential for divalent cations such as Ca^{2+} and Mg^{2+}, both of which are prevalent in seawater. However, some studies do report increasing sorption with increased salinity (Malamis and Katsou, 2013), as we observed for the quartz solid (Fig. 4), suggesting the potential for more complex underlying mechanisms. Ultimately, these results provide insight into the behavior of Ra in complex estuarine environments characterized by the presence of multiple mineral and solid forms.

4. Conclusions

Three analytical approaches were compared for rapid turnaround measurements of ^{226}Ra in simulated freshwater and seawater samples. HPGe detection following Ra extraction on an Empore™ Solid Phase Extraction (SPE) Disk had the lowest limit of detection for freshwater samples but is also the costliest measurement technique. Furthermore, there is a significant penalty to total recovery and therefore increased limits of detection for high salinity samples due to competition for sorption sites on the extraction disk with other dissolved ions. HPGe measurement of Ra also requires a greater sample volume than measurement versus LSC or ICP-MS, which represents a meaningful limitation for this method. Direct measurement of ^{226}Ra using quadrupole ICP-MS provided the second lowest detection limit for the low salinity samples. The drawbacks to ICP-MS include significant sample matrix effects as demonstrated by the suppressed sensitivity for the high salinity samples and the high cost of a traceable ^{226}Ra standard to produce the necessary calibration curve which may be prohibitive.

LSC presented the simplest method and had the lowest detection limit for high salinity samples. However, the method requires a 28-day waiting period for progeny ingrowth and the present study demonstrates that the seawater matrix has the potential to alter the consistency and coloration of scintillation fluid, which caused increased quenching and decreased detection efficiency after six months. The continuous but unpredictable degradation of both the Ra LSC standards and samples make it challenging to standardize the overall impact to detection efficiency, necessitating the frequent creation of new sets of standards (including waiting for the ingrowth period), which can be time-consuming, cumbersome, and potentially expensive. Thus, the recommended approach for accurate rapid turnaround measurements of Ra in marine waters is ICP-MS with situation-specific dilution to reduce the matrix effects. LSC may remain the preferred technique in cases of cost- or expertise-constraints, or when preservation of the samples beyond the initial ingrowth period is not needed.

CRediT authorship contribution statement

Sarah E. Donaher: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis. **Jiusi Wang:** Writing – original draft, Methodology, Investigation. **Nicole E. Martinez:** Writing – review & editing, Writing – original draft, Supervision, Resources, Conceptualization. **Brian A. Powell:** Writing – review

& editing, Writing – original draft, Visualization, Supervision, Resources, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix ASupplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.apradiso.2025.111851.

Data availability

Data will be made available on request.

References

3 M, 2000. Empore™ Radium Rad Disks: Method Summary.
Abbasi, A., 2018. A review of the analytical methodology to determine Radium-226 and Radium-228 in drinking waters. Radiochim. Acta 106 (10), 819–829. https://doi.org/10.1515/ract-2018-2967.
Acosta, J.A., Jansen, B., Kalbitz, K., Faz, A., Martínez-Martínez, S., 2011. Salinity increases mobility of heavy metals in soils. Chemosphere 85 (8), 1318–1324. https://doi.org/10.1016/j.chemosphere.2011.07.046.
Agilent Technologies, 2005. ICP-MS: Inductively Coupled Plasma Mass Spectrometry : a Primer. Agilent Technologies. https://books.google.com/books?id=gt3_OAAACAAJ.
Ames, L.L., McGarrah, J.E., Walker, B.A., Salter, P.F., 1983. Uranium and radium sorption on amorphous ferric oxyhydroxide. Chem. Geol. 40 (1–2), 135–148. https://doi.org/10.1016/0009-2541(83)90095-5.
Boudias, M., Gourgiotis, A., Montavon, G., Cazala, C., Pichon, V., Delaunay, N., 2022. 226Ra and 137Cs determination by inductively coupled plasma mass spectrometry: state of the art and perspectives including sample pretreatment and separation steps. J. Environ. Radioact. 244–245, 106812. https://doi.org/10.1016/j.jenvrad.2022.106812.
Casagrande, M.F.S., Bonotto, D.M., 2018. The use of γ-rays analysis by HPGe detector to assess the gross alpha and beta activities in waters. Appl. Radiat. Isot. 137, 1–11. https://doi.org/10.1016/j.apradiso.2018.02.027.
Currie, L.A., 1968. Limits for qualitative detection and quantitative determination. Application to radiochemistry. Anal. Chem. 40 (3), 586–593. https://doi.org/10.1021/ac60259a007.
Diego-Feliu, M., Rodellas, V., Alorda-Kleinglass, A., Tamborski, J., Van Beek, P., Heins, L., Bruach, J.M., Arnold, R., Garcia-Orellana, J., 2020. Guidelines and limits for the quantification of Ra isotopes and related radionuclides with the radium delayed coincidence counter (RaDeCC). J. Geophys. Res.: Oceans 125 (4), e2019JC015544. https://doi.org/10.1029/2019JC015544.
Esfandiar, N., Suri, R., McKenzie, E.R., 2022. Competitive sorption of Cd, Cr, Cu, Ni, Pb and Zn from stormwater runoff by five low-cost sorbents; Effects of co-contaminants, humic acid, salinity and pH. J. Hazard Mater. 423, 126938. https://doi.org/10.1016/j.jhazmat.2021.126938.
Feng, X., He, Q., Wang, J., Chen, J., 2015. The effect of sample stability on the determination of radioactivity for various radionuclides by liquid scintillation counting. Appl. Radiat. Isot. 104, 147–154. https://doi.org/10.1016/j.apradiso.2015.06.032.
Grabowski, P., Bem, H., 2010. Determination of Po and uranium in high salinity water samples. J. Radioanal. Nucl. Chem. 286 (2), 455–460. https://doi.org/10.1007/s10967-010-0720-4.
Henderson, P.B., Morris, P.J., Moore, W.S., Charette, M.A., 2013. Methodological advances for measuring low-level radium isotopes in seawater. J. Radioanal. Nucl. Chem. 296 (1), 357–362. https://doi.org/10.1007/s10967-012-2047-9.
Hou, X., 2018. Liquid scintillation counting for determination of radionuclides in environmental and nuclear application. J. Radioanal. Nucl. Chem. 318 (3), 1597–1628. https://doi.org/10.1007/s10967-018-6258-6.
IAEA, 2010. Analytical determination of radium isotopes in environmental samples. I. A. E. Agency. https://www-pub.iaea.org/MTCD/publications/PDF/IAEA-AQ-19_web.pdf.
IAEA, 2013a. Management of NORM Residues. IAEA-TECDOC-1712. International Atomic Energy Agency. https://www.iaea.org/publications/10396/management-of-norm-residues.
IAEA, 2013b. Radiation Protection and Management of NORM Residues in the Phosphate Industry. Safety Report Series No. 78. STI/PUB/1582. International Atomic Energy Agency. https://www.iaea.org/publications/8947/radiation-protection-and-management-of-norm-residues-in-the-phosphate-industry.
ICRP, 2008. Nuclear decay data for dosimetric calculations. ICRP publication 107. Ann. ICRP 38 (3).
ICRP, 2019. Radiological protection from naturally occurring radioactive material (NORM) in industrial processes. ICRP publication 142. Ann. ICRP 48 (4), 5–67. https://doi.org/10.1177/0146645319874589.
Jia, G., Jia, J., 2012. Determination of radium isotopes in environmental samples by gamma spectrometry, liquid scintillation counting and alpha spectrometry: a review of analytical methodology. J. Environ. Radioact. 106, 98–119. https://doi.org/10.1016/j.jenvrad.2011.12.003.
Knoll, G.F., 2010. Radiation Detection and Measurement, fourth ed. John Wiley.
Ku, T.-L., Mathieu, G.G., Knauss, K.G., 1977. Uranium in open ocean: concentration and isotopic composition. Deep-Sea Res. 24 (11), 1005–1017. https://doi.org/10.1016/0146-6291(77)90571-9, 1977/11/01/.
Lagacé, F., Foucher, D., Surette, C., Clarisse, O., 2017. Quantification of 226Ra at environmental relevant levels in natural waters by ICP-MS: optimization, validation and limitations of an extraction and preconcentration approach. Talanta 167, 658–665. https://doi.org/10.1016/j.talanta.2017.02.031.
MacIntosh, A., Koppel, D.J., Johansen, M.P., Beresford, N.A., Copplestone, D., Penrose, B., Cresswell, T., 2022. Radiological risk assessment to marine biota from exposure to NORM from a decommissioned offshore oil and gas pipeline. J. Environ. Radioact. 251–252, 106979. https://doi.org/10.1016/j.jenvrad.2022.106979.
Malamis, S., Katsou, E., 2013. A review on zinc and nickel adsorption on natural and modified zeolite, bentonite and vermiculite: examination of process parameters, kinetics and isotherms. J. Hazard Mater. 252–253, 428–461. https://doi.org/10.1016/j.jhazmat.2013.03.024.
McDowell, W.J., Arndsten, B.A., Case, G.N., 1989. The synergistic solvent extraction of radium from alkaline nitrate media by dicyclohexano-21-crown-7 combined with 2-methyl-2-heptyl nonanoic acid equtubrium reactions and metal ion competition. Solvent Extr. Ion Exch. 7 (3), 377–393. https://doi.org/10.1080/07360298908962314.
Nelson, A.W., May, D., Knight, A.W., Eitrheim, E.S., Mehrhoff, M., Shannon, R., Litman, R., Schultz, M.K., 2014. Matrix complications in the determination of radium levels in hydraulic fracturing flowback water from marcellus shale. Environ. Sci. Technol. Lett. 1 (3), 204–208. https://doi.org/10.1021/ez5000379.
NIST XCOM. "Element/Compound/Mixture Selection." National Institute of Standards and Technology, Physical Measurement Laboratory. https://physics.nist.gov/PhysRefData/Xcom/html/xcom1.html. Accessed 15 July 2024.
Palomo, M., Villa-Alfageme, M., Casacuberta, N., Peñalver, A., Borrull, F., Aguilar, C., 2011. Evaluation of different parameters affecting the liquid scintillation spectrometry measurement of gross alpha and beta index in water samples. Appl. Radiat. Isot. 69 (9), 1274–1281. https://doi.org/10.1016/j.apradiso.2011.04.020.
Povinec, P.P., Osvath, I., Baxter, M.S., 1996. Underwater gamma-spectrometry with HPGe and NaI(Tl) detectors. Appl. Radiat. Isot. 47 (9–10), 1127–1133. https://doi.org/10.1016/S0969-8043(96)00118-2.
Ramebäck, H., Vesterlund, A., Hedberg, M., Jonsson, S., Lagerkvist, P., Vidmar, T., 2023. Gamma spectrometric measurement of uranium isotopic composition and mass in sintered UO2 pellets using the efficiency transfer method. Appl. Radiat. Isot. 192, 110607. https://doi.org/10.1016/j.apradiso.2022.110607.
Sajih, M., Bryan, N.D., Livens, F.R., Vaughan, D.J., Descostes, M., Phrommavanh, V., Nos, J., Morris, K., 2014. Adsorption of radium and barium on goethite and ferrihydrite: a kinetic and surface complexation modelling study. Geochem. Cosmochim. Acta 146, 150–163. https://doi.org/10.1016/j.gca.2014.10.008.
Schubert, F., Kallmeyer, J., 2023. Liquid scintillation counting at the limit of detection in biogeosciences. Front. Microbiol. 14, 1194848. https://doi.org/10.3389/fmicb.2023.1194848.
Song, L.J., Yang, Y.G., Luo, M.Y., Ma, Y., Dai, X.X., 2017. Rapid determination of radium-224/226 in seawater sample by alpha spectrometry. J. Environ. Radioact. 171, 169–175. https://doi.org/10.1016/j.jenvrad.2017.01.024.
Sun, C.Q., Zhang, Y.Z., Gong, Z.S., Wang, X.J., Yang, Y., Wang, Y., 2018. Determination of trace elements in samples with high salt content by inductively coupled plasma mass spectrometry after solid-phase preconcentration. Int. J. Mass Spectrom. 431, 22–26. https://doi.org/10.1016/j.ijms.2018.04.003.
Thakur, P., Ward, A.L., González-Delgado, A.M., 2021a. Optimal methods for preparation, separation, and determination of radium isotopes in environmental and biological samples. J. Environ. Radioact. 228, 106522. https://doi.org/10.1016/j.jenvrad.2020.106522.
Thomas, R., 2004. Practical Guide to ICP-MS. Marcel Dekker, Inc. http://algol.fis.uc.pt/jap/ICP-MS-Primer-Thomas.pdf.
US EPA, 2008. Technologically Enhanced Naturally Occurring Radioactive Materials from Uranium Mining, Volume 1: Mining and Reclamation Background, Volume 2: Investigation of Potential Health, Geographic, and Environmental Issues of Abandoned Uranium Mines. US Environmental Protection Agency. *EPA 402-R-08-005*.

US EPA, 2019. High Resolution Gamma-Ray Spectrometry Analyses for Normal Operations and Radiological Incident Response. US Environmental Protection Agency. *EPA 402-B-17-001*. https://www.epa.gov/sites/default/files/2020-07/documents/guide_for_high_resolution_gamma_spectrometry_analyses_camera_ready.pdf.

US NRC, 2011. Liquid Scintillation Counting. Basic Health Physics. ML11229A718-0751-H122.

van Beek, P., Souhaut, M., Reyss, J.L., 2010. Measuring the radium quartet (^{228}Ra, ^{226}Ra, ^{224}Ra, ^{223}Ra) in seawater samples using gamma spectrometry. J. Environ. Radioact. 101 (7), 521–529. https://doi.org/10.1016/j.jenvrad.2009.12.002.

Verrezen, F., Loots, H., Hurtgen, C., 2008. A performance comparison of nine selected liquid scintillation cocktails. Appl. Radiat. Isot. 66 (6), 1038–1042. https://doi.org/10.1016/j.apradiso.2008.02.050.

Wilschefski, S.C., Baxter, M.R., 2019. Inductively coupled plasma mass spectrometry: introduction to analytical aspects. Clin. Biochem. Rev. 40 (3), 115–133. https://doi.org/10.33176/aacb-19-00024.